

LECTURE 01.

□ Course information

- ✓ Class structure
- ✓ Quizzes structure
- ✓ Grading
- ✓ Unofficial DISCORD server

pre class
class
post class

← quiz

Office Hours: M- 5-6pm
W- 9-10pm
F: 5-6pm

□ remarks:

- ✓ live proctoring
- ✓ makeup policy
- unus

13 lectures

PART 1. Introduction + modeling

• What is an ordinary differential equation?

partial DE.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

2nd order
↓
ODE

$$\dot{u}(t) + \ddot{u}(t) = u(t)$$

non-linear ODE

$$\left(\frac{du}{dt}\right)^2 + e^u = 42\pi t$$

delay DE

$$u'(t) = u(t-1)$$

Examples of ODEs. ANALYTICAL APPROACH.

$$\textcircled{1} \quad y' = t^2 + 2 \Rightarrow y(t) = \frac{t^3}{3} + 2t + C$$

$$\textcircled{2} \quad y' = \sin(t) \Rightarrow y(t) = -\cos(t) + C$$

$$\textcircled{3} \quad y'' = \frac{1}{t} \Rightarrow y' = \ln|t| + C \Rightarrow y(t) = t \ln t - t + Ct + D$$

$$\textcircled{4} \quad y' = y \Rightarrow y = 0, y(t) = Ce^t$$

$$\textcircled{5} \quad y' = ty \Rightarrow y(t) = C \cdot e^{t^2/2}, y'(t) = C \cdot e^{t^2/2} \cdot \frac{2t}{2} = t \cdot C e^{t^2/2} = t \cdot y$$

$$\textcircled{6} \frac{dy}{dt} \frac{y}{t^2} = \frac{2y}{t} + t^2 \cos t$$

$$y' \cdot \frac{1}{t^2} - \frac{2y}{t^3} = \cos t.$$

$$\left(y \cdot \frac{1}{t^2}\right)' = \cos t.$$

$$y/t^2 = \sin(t) + c \Rightarrow y(t) = t^2 \sin(t) + ct^2.$$

Initial Value Problems (IVP)

$$y' = f(t, y), \quad y(t_0) = y_0$$

$$\text{general solution: } y(t) = ct^2 + t^2 \sin(t)$$

$$\text{particular solution: } y_p(t) = t^2 \sin(t)$$

Examples:

$$\textcircled{7} \begin{cases} y' = \frac{2y}{t} + t^2 \cos(t) \\ y(\pi/2) = \frac{\pi^2}{4} \end{cases}$$

$$\textcircled{8} \begin{cases} y' = e^{t^2} \\ y(0) = 1 \end{cases}$$

$$y(t) = \cancel{ct^2} + t^2 \sin(t) = c \cdot \left(\frac{\pi^2}{2}\right) + \left(\frac{\pi}{2}\right)^2 \cdot \cancel{\sin\left(\frac{\pi}{2}\right)} = \frac{\pi^2}{4}$$

$$y\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$$

$$(c+1) \frac{\pi^2}{4} = \frac{\pi^2}{4} \Rightarrow c+1=1 \Rightarrow c=0.$$

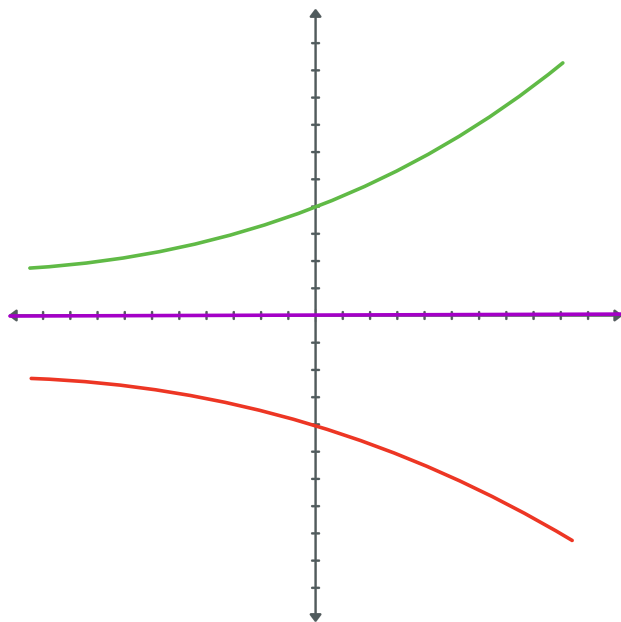
Ex. Verify that $y(t) = t^2 \sin(t)$ satisfy ⑦

MODELING

Population Growth. (unlimited resources)

$P(t)$: population at a time t .

$$\frac{dP}{dt} = k \cdot P \quad \Rightarrow \quad P(t) = C \cdot e^{kt}$$



$$P(0) = 1 \Rightarrow C = 1$$

$$P(0) = 0 \Rightarrow C = 0$$

$$P(0) = -1 \Rightarrow C = -1$$

Ex: A pop. starts at 50 (animals) and 3 years later become 150.

What is the parameter k of the model?

a) $\frac{\ln 5}{3}$ c) $\frac{\ln 3}{5}$

$$C \cdot e^{kt} \Rightarrow C = 50 = P(0).$$

~~b)~~ $\frac{\ln 3}{3}$ d) $\frac{\ln 5}{5}$

$$50 e^{k \cdot 3} = 150$$

$$e^{3k} = 3$$

$$\Rightarrow 3k = \log 3$$

$$\therefore k = \frac{\log 3}{3}$$

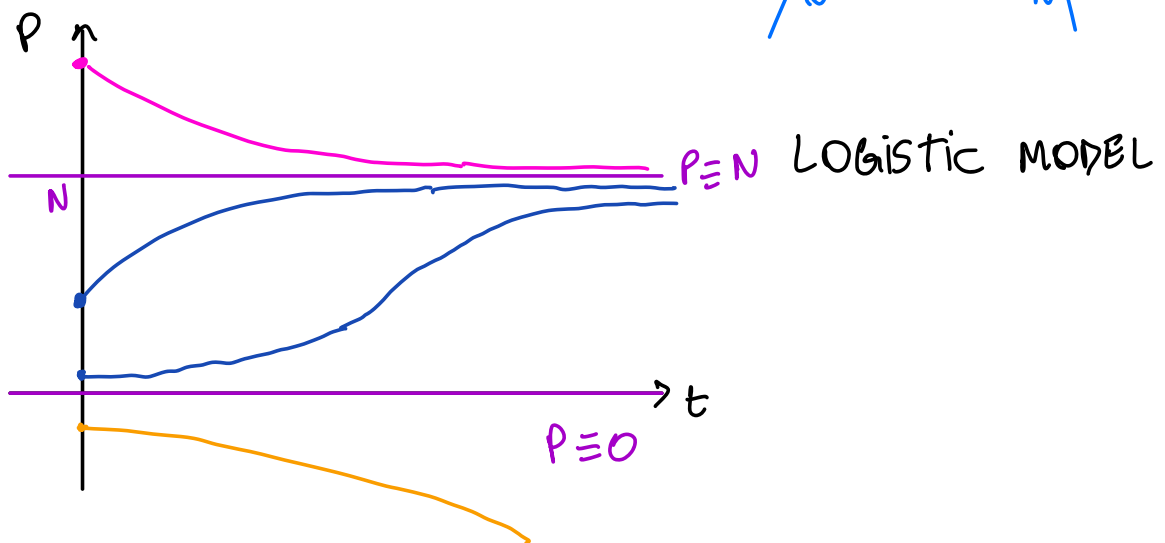
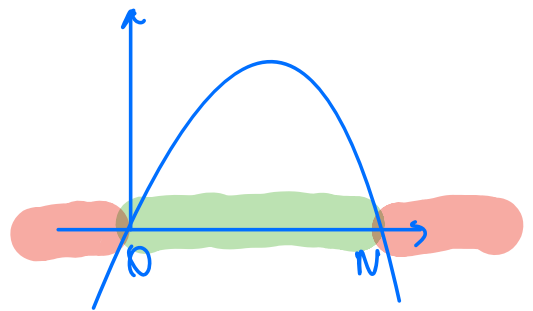
Logistic Model: A cap on the population.

carrying capacity: N

if $P \approx 0$, then $\frac{dP}{dt} \approx kP \approx 0$

$P \approx N$, then $\frac{dP}{dt} \approx 0$

$$\frac{dP}{dt} = k \cdot P \left(1 - \frac{P}{N}\right)$$



Population with harvesting:

$$\frac{dP}{dt} = \underset{\substack{\uparrow \\ \text{growth}}}{k} P \left(1 - \frac{P}{\underset{\substack{\uparrow \\ \text{carrying capacity}}}{N}}\right) - \underset{\substack{\leftarrow \\ \text{harvest parameter}}}{H}$$

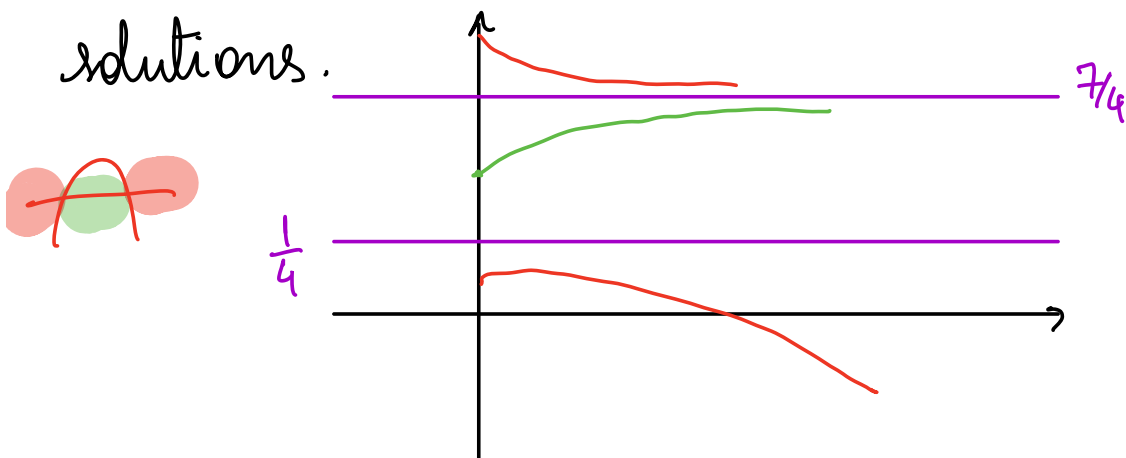
Example: Suppose

$$\frac{-4 \pm \sqrt{16 - 4 \cdot (-2) \cdot \left(-\frac{7}{8}\right)}}{2(-2)}$$

$P' = -2P^2 + 4P - \frac{7}{8} = 2P\left(1 - \frac{P}{2}\right) - \frac{7}{8}$
is a logistic model with harvest.

Find parameters and sketch graph of

solutions.



when $P' = 0$ we say that it is an equilibrium sol.

Air resistance in free-fall

Gravitational Force: mg

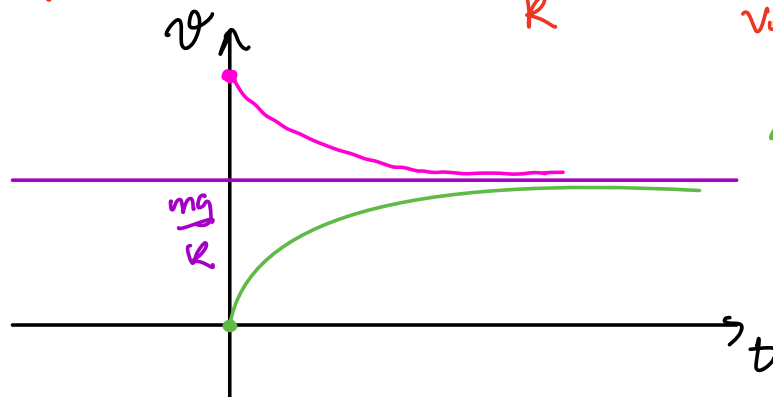
Air resistance: $-k \cdot v$

Newton's law: $m \cdot \dot{v} = mg - k \cdot v$

$$\dot{v} = g - \frac{k}{m} v = 0$$

How many equilibrium solutions?

↓ equi. sol. $\Rightarrow v_t = \frac{mg}{k}$: terminal velocity.



$$\begin{aligned}\dot{v} &= g - \frac{k}{m} \cdot 0 \\ &= g > 0\end{aligned}$$

• Predator-prey systems.

x • population of predators

y • population of prey

$$\begin{cases} \dot{x} = -Ax + Bxy \\ \dot{y} = Cy - Dxy \end{cases}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -Ax + Bxy \\ Cy - Dxy \end{bmatrix}$$

• $-Ax$ predators compete for resource.

• Bxy says predators thrive when there is a lot of interaction between prey-predator

• Cy prey thrives in isolation

• $-Dxy$ if prey dies to predator.

PART 2. Separation of variables

- Method to solve ODE analytically

- conditions: if $y' = f(t, y) = a(t) \cdot b(y)$, then that's a separable equation

- mnemonic:

$$y' = \frac{dy}{dt} = a(t)b(y) \Rightarrow dy = a(t)b(y)dt$$

$$\int \frac{dy}{b(y)} = \int a(t)dt$$

- actual math:

$$\int_{t_0}^t \frac{1}{b(y)} \frac{dy}{dt} dt = \int_{t_0}^t a(t) dt$$

$$\int_{y(t_0)}^{y(t)} \frac{1}{b(u)} du = \int_{t_0}^t a(t) dt.$$

Examples:

$$\textcircled{1}^* \quad y' = y^2 \quad a(t) = 1 \quad b(y) = y^2 \quad \int \frac{1}{y^2} dy = \int 1 dt$$

$$-\frac{1}{y} = t + C \Rightarrow -1 = y(t+C) \\ \Rightarrow y(t) = \frac{-1}{t+C}$$

Q: Do you have all the solutions?

No. $y(t) \equiv 0$ is a solution, but it is not $\frac{-1}{t+C}$ for any value of C .

$$\textcircled{2} \quad y' = \frac{y}{t^2}, \quad t > 0 \quad a(t) = \quad b(y) =$$

$$y'/y = \frac{1}{t^2} \Rightarrow \int \frac{1}{y} dy = \int \frac{1}{t^2} dt$$

$$\Rightarrow \ln|y| = -\frac{1}{t} + C$$

$$y(t) = e^{-\frac{1}{t} + C} = e^{-\frac{1}{t}} \cdot e^C = D e^{-\frac{1}{t}}$$

$y(t) \equiv 0$ is also a solution.

③ logistic model. $P' = kP\left(1 - \frac{P}{N}\right)$

$$\int \frac{P' dt}{P(1 - \frac{P}{N})} = \int k dt = kt + C$$

$$\frac{1}{P(1 - \frac{P}{N})} = \frac{\frac{1}{P}}{1} + \frac{\frac{1}{N}}{1 - \frac{P}{N}} = \frac{A - AP/N + PB}{P(1 - \frac{P}{N})} = \frac{A + (-\frac{A}{N} + B)P}{P(1 - \frac{P}{N})}$$

$$\begin{cases} A = 1 \\ -\frac{A}{N} + B = 0 \Rightarrow B = \frac{1}{N} \end{cases}$$

$$|P| = \begin{cases} P & \text{if } P \geq 0 \\ -P & \text{if } P < 0 \end{cases}$$

$$\int \frac{1}{P(1 - \frac{P}{N})} dP = \int \frac{1}{P} dP + \int \frac{1}{1 - \frac{P}{N}} dP = \ln|P| - \ln|1 - \frac{P}{N}| = \ln\left(\frac{P}{1 - \frac{P}{N}}\right)$$

$$\frac{NP}{N - P} = D \cdot e^{kt} \Rightarrow NP = (N - P)De^{kt}$$

$$NP + PDe^{kt} = NDe^{kt}$$

$$P(N + De^{kt})$$

$$P(t) = \frac{NDe^{kt}}{N + De^{kt}} \quad \left(\begin{array}{l} \text{or } P \equiv 0 \\ P \equiv N \end{array} \right)$$

if $P(0) \in (0, N)$

explicit solution: $P(t) = \dots$

④ $y' = y + t$ $a(t) =$ $b(y) =$

 this is not separable. $t^2 y + 8y = \underbrace{y}_{b(y)} \underbrace{(t^2 + 8)}_{a(t)}$
 Separation of Variables does NOT apply.

integrating factor, undetermined coefficients

Solving initial value problems:

Approach 1: Find general solution, then C.

① $\begin{cases} y' = t/y^2 \\ y(2) = 6 \end{cases} \rightarrow y^2 \frac{dy}{dt} = t \sim y^2 dy = t dt$
 $\frac{y^3}{3} = \frac{t^2}{2} + C \sim y = \left[3 \left(\frac{t^2}{2} + C \right) \right]^{1/3}$
 $y(2) = 6 = \left[3 \left(\frac{2^2}{2} + C \right) \right]^{1/3} = \left[6 + 3C \right]^{1/3}$
 $\Rightarrow 216 = 6 + 3C \Rightarrow C = 70.$

② $\begin{cases} \frac{dy}{dt} = (1+y^2)e^t \\ y(0) = \sqrt{3} \end{cases} \int \frac{dy}{1+y^2} = \int e^t dt \Rightarrow \arctan(y) = e^t + C$
 $y(t) = \tan(e^t + C)$
 $y(0) = \tan(1+C) = \sqrt{3} \Rightarrow 1+C = \frac{\pi}{3} + k\pi$
 $C = \frac{\pi}{3} - 1 + k\pi, k \in \mathbb{Z}.$

Approach 2. Implement initial condition on your integration.

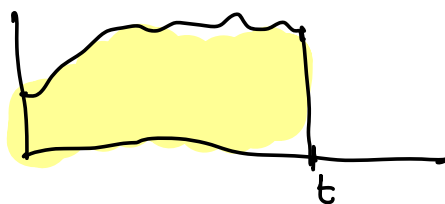
$$a) \begin{cases} \frac{dy}{dt} = \frac{1}{t \sin(y)} \\ y(e^3) = \pi \end{cases} \quad \int_{\pi}^y \sin(y) dy = \int_{e^3}^t \frac{1}{t} dt$$

$$\Rightarrow -\cos(y) + \underbrace{\cos(\pi)}_{-1} = \ln|t| - \underbrace{\ln e^3}_3$$

$$y(t) = \arccos(-\ln|t| + 2)$$

$$b) \begin{cases} \frac{dy}{dt} = e^{-y} \cdot 1 \\ y(2) = 5 \end{cases} \quad \int_5^y e^s ds = \int_2^t 1 dt \quad \cos(y) + \ln|t| - 2 = 0.$$

$$F'(t) = f(t) \quad F(t) = \int_0^t f(u) du$$



Implicit Solutions.

Ex: $\frac{dy}{dt} = \left(\frac{3y}{1-2y^3} \right) \cdot 1 \quad \leadsto \int \frac{1-2y^2}{3y} dy = t + C$

$$\frac{1}{3} \cdot \ln|y| - \frac{2}{3} \cdot \frac{1}{3} y^3 = t + C. \quad F(t, y) = 0.$$

Remark: • if the exercise is not explicit about

the form, either is fine.

• if it says explicit, then explicit it is.

Newton's Law of Cooling

Suppose an object has temperature $T(t)$, in a room of constant temperature T_0 .

There is a parameter $k > 0$, s.t.

$$\dot{T} = k(T_0 - T)$$

$$T > T_0 \rightarrow \dot{T} < 0$$

$$T < T_0 \rightarrow \dot{T} > 0$$

Ex: A cup of coffee is at 200° initially in a room of temperature 70° .

After 10 minutes, it is at 150° .

1) How long until it reaches 100° ?

a) 10 min

b) 20 min

~~c) 30 min~~

d) 40 min

2) What about 70° ?

a) 20 min

b) 60 min

c) 120 min

d) never

IVP.

$$\begin{cases} T' = k(70 - T) \\ T(0) = 200 \end{cases}$$

$$\int_{200}^T \frac{dT}{70 - T} = \int_0^t k dt = kt$$

$$-\ln|70 - T| + \ln|70 - 200| = kt$$

$$\ln\left|\frac{130}{70 - T}\right| = kt \quad \hookrightarrow \frac{130}{70 - T} = e^{kt}$$

$$k = \frac{1}{10} \cdot \ln\left(\frac{13}{8}\right) > 0$$

$$\frac{130}{100 - 70} = e^{k \cdot t}$$

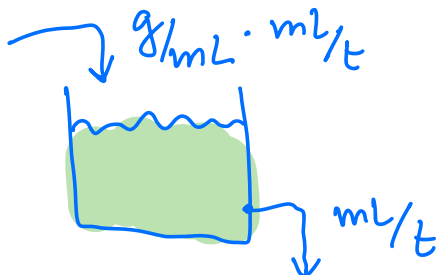
$$\frac{13}{3} = e^{k \cdot t}$$

$$t = \frac{1}{k} \cdot \ln\left(\frac{13}{8}\right)$$

Mixing Problem

A chemical goes in with some fixed concentration, and water comes out at some rate.

Q: concentration of chemical at any given time.



Ex: A lake has volume $30 km^3$, initially with

$5 kg/km^3$ of salt. Water containing $12 kg/km^3$

flows in at a rate of $3 km^3$ per month. Water flows out at a rate of $3 km^3$ per month. When will the concentration reach $10 kg/km^3$? Assume $\ln(7/2) \approx 1.25$.

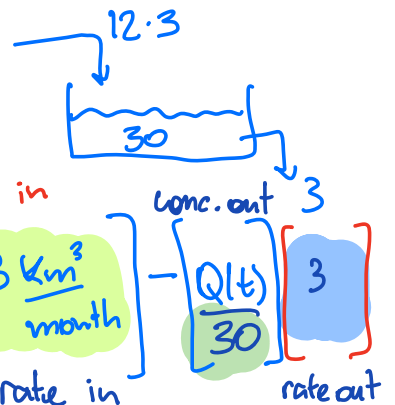
- a) 13 months
- b) 7 months
- c) 8 months
- d) 10 months

$$V(t) = 30.$$

$$Q(0) = 5$$

$$Q'(t) = \left[\frac{12 \cdot kg}{km^3} \right] \left[\frac{3 km^3}{month} \right] - \left[\frac{Q(t)}{30} \right] \left[3 \right]$$

concentration in
conc. out
rate in
rate out



$Q'(t) = \text{amount in} - \text{amount out}$
 density · volume density · volume.

$$Q'(t) = 36 - \frac{Q}{10} \rightsquigarrow \frac{dQ}{36 - \frac{Q}{10}} = 1 dt$$

$$Q(0) = 5$$

- 1st order differential eq.

$$y' = f(t, y)$$

What is the order of ODE?

A: number of the highest derivative.

$$y^{(k)} = f(t, y, \dots, y^{(k-1)}) \quad k\text{-th order ODE.}$$

- $y' = \underbrace{a(t)} \cdot y + \underbrace{b(t)}$ $y = ax + b$

ODE is linear. otherwise, non-linear.

—h—

END · Lecture 01.

Preclass 2
Postclass 1

HW 1-4
Sections 1.1, 1.2

Fri { 20-25 min Quiz
Part 1
— break
Part 2.